PHYS4055

Atomic Physics and Quantum Optics

Worked Examples and Exam Methods

Four-part companion to the course overview

2026 Open-book exam revision

What this booklet does. Each example starts with a recognisable tutorial, assignment or past-exam question type and then shows the physics model, the algebra, the final result and at least one independent check. The source is identified with each example.

Course part	Core worked problems
I. Quantum optics	Displacement and coherent states; two-mode squeezed vacuum; beam splitters, HOM/NOON states; photon statistics and squeezing
II. Applications of optics	Bell pairs and dense coding; BB84; distillation; Kraus loss; conditional filtering; Tavis–Cummings; cavity squeezing; metrology
III. Atomic physics	Finite-nuclear-size correction; equivalent d^2 electrons; Stark polarizability; hyperfine/Zeeman; direct/exchange Hartree–Fock
IV. Atomic applications	E1/M1/E2 selection rules and lifetimes; Yb laser cooling and Zeeman slower; cesium clock and parity violation

Verification status. Examples cross-reference original course material. The Optics Assignment 1 and 2 questions have released solutions, whereas the atomic assignment questions do not all have official solutions. These sections give independently derived study solutions, not a substitute for unpublished marking schemes. Any exact values labelled *illustrative* are not claimed as experimental data. The original PDFs remain authoritative for handwritten symbols and diagrams.

Updated 13 November 2026. All calculations use explicitly stated sign/normalisation conventions.

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Part I – Quantum optics

1 QO-01: Displacement operators and coherent-state fields

Course source: Week 1 Tutorial, Questions 1–3; `Week1Solutions (1).pdf`.

Recognise the problem

A coherent state is a displaced vacuum. The key identity $\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha$ lets us calculate its observables without expanding infinitely many number states.

Step 1: evaluate the commutator

Write $\hat{A} = \alpha\hat{a}^\dagger - \alpha^*\hat{a}$, so $\hat{D} = e^{\hat{A}}$ and $\hat{A}^\dagger = -\hat{A}$. Using the Baker–Campbell–Hausdorff identity,

$$e^{-\hat{A}}\hat{a}e^{\hat{A}} = \hat{a} + [-\hat{A}, \hat{a}] + \frac{1}{2!}[-\hat{A}, [-\hat{A}, \hat{a}]] + \cdots .$$

Since $[\hat{a}^\dagger, \hat{a}] = -1$,

$$[-\hat{A}, \hat{a}] = -\alpha[\hat{a}^\dagger, \hat{a}] + \alpha^*[\hat{a}, \hat{a}] = \alpha.$$

This is a scalar, so every further nested commutator vanishes:

$$\boxed{\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha.}$$

Step 2: show that a coherent state is an eigenstate

Defining $|\alpha\rangle = \hat{D}(\alpha)\,|0\rangle$,

$$\hat{a}\,|\alpha\rangle = \hat{D}(\alpha)(\hat{D}^\dagger\hat{a}\hat{D})\,|0\rangle = \hat{D}(\alpha)(\hat{a} + \alpha)\,|0\rangle = \boxed{\alpha\,|\alpha\rangle}.$$

Therefore $\langle\hat{a}\rangle_\alpha = \alpha$ and $\langle\hat{n}\rangle = |\alpha|^2$.

Step 3: recover a classical oscillating field

For the Week 1 travelling-wave operator,

$$\hat{\mathbf{E}}(z,t) = \mathbf{e}E_0[\hat{a}e^{i\varphi} + \hat{a}^\dagger e^{-i\varphi}], \quad \varphi = kz - \omega t,$$

write $\alpha = |\alpha|e^{i\theta}$. Then

$$\langle\hat{\mathbf{E}}\rangle_\alpha = \mathbf{e}E_0(\alpha e^{i\varphi} + \alpha^* e^{-i\varphi}) = \boxed{2\mathbf{e}E_0|\alpha|\cos(kz - \omega t + \theta)}.$$

> **Physical check / common trap.** The result must be real and oscillate at the mode frequency. A number state instead has $\langle\hat{a}\rangle = 0$; nonzero average energy alone does not yield a classical mean field.

2 QO-02: Two-mode squeezed vacuum, correlations and heralding

Course source: 2026 Optics Assignment 1, Question 1; released Assignment 1 solutions.

Given state and first checks

Let $\lambda = \tanh r$, $0 \le \lambda < 1$, and

$$|\Psi\rangle = \sqrt{1-\lambda^2}\sum_{n=0}^{\infty}\lambda^n\,|n\rangle_s\,|n\rangle_i\,.$$

Normalisation follows from $(1-\lambda^2)\sum_{n=0}^{\infty}(\lambda^2)^n = 1$. Every pair has identical signal and idler photon number.

Step 1: trace out the idler

Expand $\rho_{si} = |\Psi\rangle\langle\Psi|$:

$$\rho_{si} = (1-\lambda^2)\sum_{m,n} \lambda^{m+n} |m\rangle_s \langle n|_s \otimes |m\rangle_i \langle n|_i .$$

The idler trace gives $\mathrm{Tr}_i(|m\rangle\langle n|) = \delta_{mn}$:

$$\boxed{\rho_s = (1-\lambda^2)\sum_{n=0}^{\infty} \lambda^{2n} |n\rangle\langle n|}.$$

This is thermal photon statistics: set $q = \lambda^2$ so $p_n = (1-q)q^n$. Differentiating the geometric series,

$$\bar{n} = (1-q)\sum_n nq^n = \frac{q}{1-q} = \boxed{\frac{\lambda^2}{1-\lambda^2} = \sinh^2 r}.$$

The *joint* state is pure, but either subsystem is mixed.

Step 2: calculate the signal–idler correlation

Because $n_s = n_i = n$ in every term,

$$\langle \hat{n}_s \hat{n}_i \rangle = \langle n^2 \rangle, \qquad \langle n^2 \rangle = (\Delta n)^2 + \bar{n}^2 = (\bar{n} + \bar{n}^2) + \bar{n}^2.$$

Hence

$$\boxed{g_{si}^{(2)}(0) = \frac{\langle \hat{n}_s \hat{n}_i \rangle}{\bar{n}^2} = 2 + \frac{1}{\bar{n}}}.$$

For example $\lambda = 0.2$ gives $\bar{n} = 1/24 \simeq 0.0417$ and $g_{si}^{(2)} = 26$.

Step 3: explain heralding

At low gain $\lambda \ll 1$, $|\Psi\rangle \simeq |0,0\rangle + \lambda |1,1\rangle + O(\lambda^2)$. Conditioning on detecting exactly one idler photon projects the signal onto $|1\rangle$. A threshold detector with nonzero multi-pair probability does not guarantee a perfect $|1\rangle$ at higher gain.

> **Physical check / common trap.** A reduced thermal state does *not* contradict strong pairwise correlations: the traced-out idler held the information about the signal photon number.

3 QO-03: Beam splitters, HOM interference and NOON states

Course source: Week 2 Tutorial, Questions 1–2; Optics Assignment 1 Question 2; 2025 Exam Question 3; Week 3 Tutorial worked sheet.

Step 1: select one convention and keep it

Use the balanced transformation from the past exams, written in terms of input creation operators,

$$\boxed{\hat{a}^\dagger \longrightarrow \frac{\hat{c}^\dagger + \hat{d}^\dagger}{\sqrt{2}}, \qquad \hat{b}^\dagger \longrightarrow \frac{\hat{c}^\dagger - \hat{d}^\dagger}{\sqrt{2}}}.$$

The four-port device conserves photon number because the mode matrix is unitary. For a general real splitter with $t^2 + r^2 = 1$, $a^\dagger \to tc^\dagger + rd^\dagger$,

$$|n,0\rangle \longrightarrow \sum_{k=0}^{n} \sqrt{\binom{n}{k}}\, t^k r^{n-k} |k, n-k\rangle .$$

Thus $P(k, n-k) = \binom{n}{k} t^{2k} r^{2(n-k)}$, the familiar binomial distribution.

Step 2: expand each two-photon input

Use $|2,0\rangle = (a^\dagger)^2 |0\rangle / \sqrt{2}$. The normalised output states are

$$\begin{aligned} |2,0\rangle_{ab} &\longrightarrow \tfrac{1}{2}|2,0\rangle_{cd} + \tfrac{1}{\sqrt{2}}|1,1\rangle_{cd} + \tfrac{1}{2}|0,2\rangle_{cd}, \\ |0,2\rangle_{ab} &\longrightarrow \tfrac{1}{2}|2,0\rangle_{cd} - \tfrac{1}{\sqrt{2}}|1,1\rangle_{cd} + \tfrac{1}{2}|0,2\rangle_{cd}, \\ |1,1\rangle_{ab} &\longrightarrow \frac{|2,0\rangle_{cd} - |0,2\rangle_{cd}}{\sqrt{2}}. \end{aligned}$$

The last line is the Hong–Ou–Mandel result: indistinguishable photons entering opposite ports never produce a coincidence.

Step 3: solve the 2025 NOON input

For $|\Psi_{\text{in}}\rangle = (|2,0\rangle + |0,2\rangle)/\sqrt{2}$, the opposite-sign $|1,1\rangle$ amplitudes cancel:

$$\boxed{|\Psi_{\text{out}}\rangle = \frac{|2,0\rangle + |0,2\rangle}{\sqrt{2}}, \quad P_{20} = P_{02} = \tfrac{1}{2}, \quad P_{11} = 0.}$$

The plus NOON state is invariant under this particular balanced beam splitter. The *minus* NOON input would instead become $|1,1\rangle$.

For the alternative 2025 input $|\Phi_{\text{in}}\rangle = (|1,1\rangle + |2,0\rangle)/\sqrt{2}$, only the second component creates $|1,1\rangle_{cd}$: its amplitude is $(1/\sqrt{2})(1/\sqrt{2}) = 1/2$, so

$$\boxed{P_{11} = \left|\tfrac{1}{2}\right|^2 = \tfrac{1}{4}.}$$

Physical check / common trap. Always convert powers of creation operators back into *normalised* Fock states; the $\sqrt{2!}$ factors change probabilities. Compare amplitudes before squaring them.

4 QO-04: Photon statistics, quadrature noise and squeezing

Course source: Week 3 Tutorial, 2024 Exam Question 3; Weeks 1–3 lecture slides.

Step 1: a two-number-state superposition

For $|\psi\rangle = \alpha|n\rangle + \beta|n+1\rangle$, $|\alpha|^2 + |\beta|^2 = 1$, put $p = |\beta|^2$. Then

$$\langle\hat{n}\rangle = n + p, \quad \langle\hat{n}^2\rangle = n^2 + (2n+1)p,$$

$$\boxed{\mathrm{Var}(\hat{n}) = p(1-p)}.$$

For any non-vacuum mean, this is less than $\bar{n} = n+p$ except the trivial vacuum limit: the statistics are sub-Poissonian. Contrast coherent light $\mathrm{Var}(n) = \bar{n}$ and single-mode thermal light $\mathrm{Var}(n) = \bar{n} + \bar{n}^2$.

Step 2: set the quadrature convention

The lectures use

$$\hat{x} = \hat{a} + \hat{a}^\dagger, \qquad \hat{p} = i(\hat{a}^\dagger - \hat{a}).$$

Since $[\hat{x}, \hat{p}] = 2i$, Heisenberg's inequality gives $(\Delta x)^2(\Delta p)^2 \geq 1$. A coherent state has $\langle x\rangle = 2\,\mathrm{Re}\,\alpha$, $\langle p\rangle = 2\,\mathrm{Im}\,\alpha$, and both variances equal 1.

Step 3: use the squeezing transformation

For real squeezing $r > 0$, let

$$\hat{S}(r) = \exp\left[\frac{r}{2}(\hat{a}^2 - \hat{a}^{\dagger 2})\right], \quad \hat{S}^\dagger \hat{a} \hat{S} = \hat{a}\cosh r - \hat{a}^\dagger \sinh r.$$

Consequently

$$\hat{S}^\dagger \hat{x} \hat{S} = e^{-r}\hat{x}, \qquad \hat{S}^\dagger \hat{p} \hat{S} = e^{r}\hat{p}.$$

Evaluating in the vacuum gives the squeezed-vacuum result

$$\boxed{(\Delta x)^2 = e^{-2r},\ (\Delta p)^2 = e^{2r},\ (\Delta x)^2(\Delta p)^2 = 1.}$$

For $r = \frac{1}{2}\ln 2$, the amplitude-quadrature variance is $1/2$, the phase-quadrature variance is 2, and the product still saturates the uncertainty bound.

Physical check / common trap. Squeezing redistributes quantum noise; it never violates the uncertainty relation. Some textbooks use $x = (a + a^\dagger)/\sqrt{2}$; first check the vacuum variance.

Part II – Applications of optics

5 OA-01: Bell states, partial traces and dense coding

Course source: 2026 Optics Assignment 2, Questions 1–2; released solutions.

Step 1: build the joint density matrix

For $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$,

$$\rho_{AB} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|).$$

Tracing over Bob uses $\mathrm{Tr}_B(|i,j\rangle\langle k,l|) = \delta_{jl}|i\rangle\langle k|$. The cross terms disappear:

$$\boxed{\rho_A = \mathrm{Tr}_B\,\rho_{AB} = \tfrac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \tfrac{1}{2}I}.$$

Thus Alice's measurement is 0 or 1 with probability 1/2, even though the two-qubit state is pure. $\mathrm{Tr}\,\rho_A^2 = \frac{1}{2} < 1$ confirms mixing.

Step 2: rewrite the correlations in another basis

With $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$, substitute $|0\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$, $|1\rangle = (|+\rangle - |-\rangle)/\sqrt{2}$. The cross terms cancel:

$$\boxed{|\Phi^+\rangle = \frac{|{+}{+}\rangle + |{-}{-}\rangle}{\sqrt{2}}}.$$

The outcomes are individually random but perfectly correlated in both the Z and X bases.

Step 3: superdense coding

Alice and Bob already share $|\Phi^+\rangle$. Alice acts on *her* qubit:

Message	Alice's gate	shared state
00	I	$\|\Phi^+\rangle$
01	X	$\|\Psi^+\rangle$
10	Z	$\|\Phi^-\rangle$
11	XZ	$-\|\Psi^-\rangle$

The last minus is an irrelevant global phase. The four Bell states are mutually orthogonal: e.g., $\langle\Phi^+|\Phi^-\rangle = (1-1)/2 = 0$. After Alice sends her photon, an ideal Bell-basis measurement at Bob identifies two classical bits.

Step 4: why ordinary linear optics is limited

At a passive 50:50 splitter, bosonic interference makes Ψ^- and Ψ^+ distinguishable using appropriate port/polarisation coincidence patterns, but Φ^+ and Φ^- share the same unresolved patterns under the standard passive scheme. Consequently only two of four equally likely Bell messages are unambiguously identified:

$$\boxed{P_{\text{unambiguous}} = \tfrac{2}{4} = \tfrac{1}{2}}.$$

This is a measurement limitation, not a failure of dense coding with an ideal Bell measurement.

> **Physical check / common trap.** A reduced mixed state alone does not prove that the joint state is mixed. The two independent random local strings can still be strongly correlated.

6 OA-02: BB84 basis sifting and intercept–resend

Course source: 2026 Optics Assignment 2, Question 3; released solutions.

Step 1: record the encoding

Use $Z = \{|H\rangle, |V\rangle\}$ and $X = \{|D\rangle, |A\rangle\}$, with $|D\rangle = (|H\rangle + |V\rangle)/\sqrt{2}$ and $|A\rangle = (|H\rangle - |V\rangle)/\sqrt{2}$.

Bit	Z basis	X basis
0	$\|H\rangle$	$\|D\rangle$
1	$\|V\rangle$	$\|A\rangle$

Alice and Bob choose bases independently and uniformly, so they match with probability 1/2.

Step 2: account for transmission and detection

For N transmitted pulses, channel transmissivity η_{ch} and detector efficiency η_d, assuming single photons and no dark counts,

$$\boxed{\mathbb{E}[N_{\text{detected}}] = N\eta_{\text{ch}}\eta_d}, \qquad \boxed{\mathbb{E}[N_{\text{sifted}}] = \tfrac{1}{2}N\eta_{\text{ch}}\eta_d}.$$

For example, with $N = 10^6$, $\eta_{\text{ch}} = 0.2$, $\eta_d = 0.8$, expect 160,000 detected and 80,000 sifted bits *before* sampling and key distillation.

Step 3: derive the intercept–resend error rate

Condition on an Alice–Bob matched basis (the sifted key):

$$P(\text{Eve matches Alice}) = \tfrac{1}{2}, \quad P(\text{error} \mid \text{match}) = 0.$$

If Eve chose the wrong conjugate basis, her result is random; once Bob measures in Alice's basis, he has a 1/2 chance of the wrong bit. Therefore

$$\boxed{\text{QBER} = \tfrac{1}{2}(0) + \tfrac{1}{2}(\tfrac{1}{2}) = \tfrac{1}{4}}.$$

To estimate the QBER, they publicly reveal m sifted bits. Under independent errors,

$$P(\geq 1 \text{ error observed}) = 1 - (1 - Q)^m.$$

For $m = 20,\ Q = 1/4$,

$$\boxed{P(\text{detect attack}) = 1 - (3/4)^{20} \simeq 0.99683}.$$

Those 20 disclosed bits must be removed from the secret key. Error correction reconciles discrepancies; privacy amplification reduces an eavesdropper's possible information.

> **Physical check / common trap.** The 25% is calculated *after basis sifting*, not across all sent photons. The 50% sift factor is independent of optical transmission in this ideal model.

7 OA-03: One round of entanglement distillation

Course source: Week 4 Tutorial, Question 1 (`Week4TutorialQuestion (1).pdf`).

Set up a noisy pair

Alice and Bob share

$$\rho = p\left|\Phi^+\right\rangle\left\langle\Phi^+\right| + (1-p)\left|\Psi^+\right\rangle\left\langle\Psi^+\right|, \qquad \tfrac{1}{2} < p \leq 1.$$

The input fidelity to $|\Phi^+\rangle$ is $F_{\text{in}} = p$. Take two independent copies. Each party performs a CNOT from their *first* pair to their *second* pair, then measures the target pair and keeps the control pair only if the target measurement results agree.

Step 1: classify the four pure branches

The four possibilities and their probabilities are

Control $\otimes$ target	Weight	Target results agree?
$\Phi^+ \otimes \Phi^+$	p^2	yes
$\Phi^+ \otimes \Psi^+$	$p(1-p)$	no
$\Psi^+ \otimes \Phi^+$	$(1-p)p$	no
$\Psi^+ \otimes \Psi^+$	$(1-p)^2$	yes

The bilateral CNOT copies the *bit-flip error label* of the two Bell pairs into the target pair: matching labels lead to equal target outcomes. For example, $|\Phi^+\rangle|\Phi^+\rangle$ remains two Φ^+ pairs, whereas $|\Phi^+\rangle|\Psi^+\rangle$ produces a target Ψ^+, whose two computational outcomes differ.

Step 2: condition on successful heralding

The successful event has probability

$$\boxed{P_{\text{suc}} = p^2 + (1-p)^2}.$$

The (unnormalised) retained control state is

$$\tilde{\rho} = p^2 \left|\Phi^+\right\rangle\left\langle\Phi^+\right| + (1-p)^2 \left|\Psi^+\right\rangle\left\langle\Psi^+\right|.$$

Normalise by its trace:

$$\boxed{\rho' = \frac{p^2}{p^2+(1-p)^2}\left|\Phi^+\right\rangle\left\langle\Phi^+\right| + \frac{(1-p)^2}{p^2+(1-p)^2}\left|\Psi^+\right\rangle\left\langle\Psi^+\right|}.$$

Therefore

$$\boxed{F' = \frac{p^2}{p^2+(1-p)^2} > p \quad \text{for } 1/2 < p < 1}.$$

For $p = 0.8$, $P_{\text{suc}} = 0.68$ and $F' = 0.64/0.68 \simeq 0.9412$.

Interpretation

The fidelity improves *conditional on success*, but each round consumes two pairs to create at most one. Repeated rounds improve quality at the expense of yield.

> **Physical check / common trap.** Postselection requires two distinct quantities: the trace (success probability) and the normalised postselected density matrix. Do not confuse them.

8 OA-04: Pure-loss channel, Kraus operators and purity

Course source: Week 5 Tutorial, Question 1; `Week5Solutions (1).pdf`.

Step 1: write a joint unitary on a single-rail qubit

Let $|\psi\rangle_S = \alpha|0\rangle + \beta|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$, and start the environment in vacuum. The loss model is

$$|0,0\rangle \mapsto |0,0\rangle, \quad |1,0\rangle \mapsto \sqrt{\eta}\,|1,0\rangle + \sqrt{1-\eta}\,|0,1\rangle.$$

Thus

$$|\Psi_{\text{out}}\rangle = \alpha|0,0\rangle + \beta\sqrt{\eta}\,|1,0\rangle + \beta\sqrt{1-\eta}\,|0,1\rangle.$$

Step 2: identify Kraus operators

Project the environment onto $\langle 0|_E$ and $\langle 1|_E$:

$$K_0 = |0\rangle\langle 0| + \sqrt{\eta}\,|1\rangle\langle 1|, \qquad K_1 = \sqrt{1-\eta}\,|0\rangle\langle 1|.$$

Explicitly,

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{\eta} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{1-\eta} \\ 0 & 0 \end{pmatrix}, \quad K_0^\dagger K_0 + K_1^\dagger K_1 = I.$$

Step 3: trace over the unobserved environment

The physical output is $\rho' = K_0 \rho K_0^\dagger + K_1 \rho K_1^\dagger$:

$$\boxed{\rho' = \begin{pmatrix} |\alpha|^2 + (1-\eta)|\beta|^2 & \sqrt{\eta}\alpha\beta^* \\ \sqrt{\eta}\alpha^*\beta & \eta|\beta|^2 \end{pmatrix}}.$$

The number population falls by η, but vacuum–one-photon coherence only by $\sqrt{\eta}$. The Kraus branch weights are

$$w_0 = |\alpha|^2 + \eta|\beta|^2, \quad w_1 = (1-\eta)|\beta|^2, \quad w_0 + w_1 = 1.$$

The environment has not been measured, so these branches are *added*, not individually normalised.

Step 4: calculate purity

For a 2×2 density matrix with trace 1, $\mathrm{Tr}(\rho'^2) = 1 - 2\det\rho'$. Here $\det\rho' = \eta(1-\eta)|\beta|^4$, hence

$$\boxed{\mathrm{Tr}(\rho'^2) = 1 - 2\eta(1-\eta)|\beta|^4}.$$

For an input $|1\rangle$, $P(1) = \eta,\ P(0) = 1-\eta$, $\langle n\rangle = \langle n^2\rangle = \eta$, $\mathrm{Var}(n) = \eta(1-\eta)$, greatest at $\eta = 1/2$.

Physical check / common trap. Unitary evolution of system+environment can be pure while the reduced system becomes mixed. Trace and positivity are your first checks.

9 OA-05: Conditional filtering can distinguish non-orthogonal states

Course source: Week 6 Tutorial, Question 2; released Week 6 tutorial solutions.

Step 1: measure the input overlap

The inputs and successful Kraus operator are

$$|\psi_\pm\rangle = \frac{2|0\rangle \pm |1\rangle}{\sqrt{5}}, \qquad K_s = \tfrac{1}{2}|0\rangle\langle 0| + |1\rangle\langle 1|.$$

Both input states are normalised, but

$$\boxed{\langle\psi_+|\psi_-\rangle = \frac{4-1}{5} = \frac{3}{5} \neq 0}.$$

Thus no *deterministic* measurement can perfectly distinguish them.

Step 2: apply the success operator before normalising

$$K_s|\psi_\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{5}}.$$

The trace/weight of this unnormalised branch is

$$\boxed{p_s = \|K_s|\psi_\pm\rangle\|^2 = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}},$$

the same for either input. Conditional on success,

$$\boxed{|\phi_\pm\rangle = \frac{K_s|\psi_\pm\rangle}{\sqrt{p_s}} = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}}, \quad \langle\phi_+|\phi_-\rangle = 0.$$

A subsequent measurement in the $\{|+\rangle, |-\rangle\}$ basis identifies which input was sent *when the filter succeeds.*

Step 3: explain the failure outcome

The other Kraus operator is $K_f = (\sqrt{3}/2)|0\rangle\langle 0|$. Check completeness:

$$K_s^\dagger K_s + K_f^\dagger K_f = \begin{pmatrix} 1/4 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 3/4 & 0 \\ 0 & 0 \end{pmatrix} = I.$$

Each failed branch is

$$K_f|\psi_\pm\rangle = \sqrt{\frac{3}{5}}|0\rangle, \quad p_f = \frac{3}{5}, \quad |\phi_f\rangle = |0\rangle.$$

Failure produces the *same* state for both signs and reveals no information distinguishing the inputs.

Physical check / common trap. The output states become orthogonal only in the successful subensemble. It is the non-unit success probability that prevents violation of deterministic distinguishability.

10 OA-06: Tavis–Cummings bright and dark states

Course source: Week 6 Tutorial, Question 1; released Week 6 tutorial solutions.

Step 1: identify the coupled subspace

For two identical resonant atoms and one cavity mode,

$$H_I = \hbar g(aJ_+ + a^\dagger J_-), \qquad J_- = \sigma_-^{(1)} + \sigma_-^{(2)}.$$

Define $|S\rangle = (|eg\rangle + |ge\rangle)/\sqrt{2}$ and $|A\rangle = (|eg\rangle - |ge\rangle)/\sqrt{2}$. Act on each state:

$$J_- |S\rangle = \sqrt{2}\,|gg\rangle\,, \qquad J_- |A\rangle = 0.$$

The symmetric state is *bright* with enhanced coupling $\sqrt{2}g$; the antisymmetric state is *dark*.

Step 2: propagate a localised excitation

Initially $|\Psi(0)\rangle = |eg;0\rangle = (|S;0\rangle + |A;0\rangle)/\sqrt{2}$. Set $\vartheta = \sqrt{2}gt$. The bright component Rabi-oscillates to $|gg;1\rangle$ while the dark component does not:

$$|\Psi(t)\rangle = \frac{1}{\sqrt{2}}\left[\cos\vartheta\,|S;0\rangle + |A;0\rangle - i\sin\vartheta\,|gg;1\rangle\right].$$

Return to the individual-atom basis:

$$\boxed{|\Psi(t)\rangle = \frac{1+\cos\vartheta}{2}\,|eg;0\rangle + \frac{\cos\vartheta - 1}{2}\,|ge;0\rangle - \frac{i\sin\vartheta}{\sqrt{2}}\,|gg;1\rangle\,.}$$

Step 3: calculate observables and conserved excitation

The cavity has exactly one photon in the last branch:

$$\boxed{\langle n_{\text{cav}}\rangle = \tfrac{1}{2}\sin^2\vartheta}.$$

Atom 1 is excited only in the first term, atom 2 only in the second:

$$P(e_1) = \frac{(1+\cos\vartheta)^2}{4}, \qquad P(e_2) = \frac{(1-\cos\vartheta)^2}{4}.$$

Thus $\langle N_{\text{at}}\rangle = P(e_1) + P(e_2) = \frac{1}{2}(1+\cos^2\vartheta)$ and

$$\boxed{\langle N_{\text{at}}\rangle + \langle n_{\text{cav}}\rangle = 1}.$$

At $\vartheta = \pi$, the excitation transfers entirely from atom 1 to atom 2 (up to phase). But $P(n_{\text{cav}} = 1) \le 1/2$: the dark half cannot populate the cavity.

> **Physical check / common trap.** Writing the initial state in the bright/dark basis is the key shortcut. Conservation of one excitation is a useful error check at every time.

11 OA-07: Cavity squeezing from a Langevin equation

Course source: 2025 PHYS4055 Exam, Question 4(a)–(d); Week 5 open-systems lectures.

Step 1: obtain the parametric interaction term

The interaction Hamiltonian and bare cavity equation are

$$H_I = \frac{i\hbar\chi}{2}(a^2 - a^{\dagger 2}), \quad \dot{a}\big|_{\chi=0} = -\tfrac{\gamma}{2}a + \sqrt{\gamma}A_{\text{in}}.$$

Using $[a^{\dagger 2}, a] = -2a^\dagger$,

$$\frac{i}{\hbar}[H_I, a] = \frac{i}{\hbar}\left(\frac{i\hbar\chi}{2}[\,a^2 - a^{\dagger 2}, a]\right) = -\chi a^\dagger.$$

Therefore

$$\boxed{\dot{a} = -\chi a^\dagger - \tfrac{\gamma}{2}a + \sqrt{\gamma}A_{\text{in}}}.$$

Step 2: transform the measured quadrature

Define $X_+ = a + a^\dagger$, and the corresponding input $X_{+,\mathrm{in}} = A_{\mathrm{in}} + A_{\mathrm{in}}^\dagger$. Adding the equation to its adjoint gives

$$\boxed{\dot{X}_+ = -(\gamma/2 + \chi)X_+ + \sqrt{\gamma}X_{+,\mathrm{in}}}.$$

Take the Fourier convention $\tilde{X}(\omega) = \int_{-\infty}^{\infty} e^{-i\omega t}X(t)\,dt$, so $\widetilde{\dot{X}} = i\omega\tilde{X}$. Then

$$\boxed{\tilde{X}_+(\omega) = \frac{\sqrt{\gamma}}{\gamma/2 + \chi + i\omega}\tilde{X}_{+,\mathrm{in}}(\omega)}.$$

Step 3: apply input–output theory

With the convention in the exam, $A_{\mathrm{out}} = \sqrt{\gamma}\,a - A_{\mathrm{in}}$, so $X_{+,\mathrm{out}} = \sqrt{\gamma}X_+ - X_{+,\mathrm{in}}$. Consequently

$$\boxed{\tilde{X}_{+,\mathrm{out}} = \frac{\gamma/2 - \chi - i\omega}{\gamma/2 + \chi + i\omega}\,\tilde{X}_{+,\mathrm{in}}}.$$

For a vacuum input with unit noise spectral density and stable dynamics, the output spectral noise is

$$\boxed{V_+(\omega) = \frac{(\gamma/2 - \chi)^2 + \omega^2}{(\gamma/2 + \chi)^2 + \omega^2}}.$$

For $0 < \chi < \gamma/2$, $V_+ < 1$ at any finite frequency, approaching 1 at high frequency. At $\omega = 0$, $V_+(0) = [(\gamma/2 - \chi)/(\gamma/2 + \chi)]^2$. The conjugate quadrature is anti-squeezed.

Physical check / common trap. State the Fourier convention and output-field sign before doing algebra; otherwise phase factors can appear inconsistent. A *spectral variance* is based on fluctuation correlations, not the squared mean field.

12 OA-08: Phase estimation with Fisher information

Course source: Week 6 Lecture 3, Quantum Optical Metrology; Mach–Zehnder slides.

Step 1: identify the phase generator

If a pure probe state is encoded by $|\psi_\phi\rangle = e^{-i\phi G}\,|\psi_0\rangle$, the quantum Fisher information is

$$\boxed{F_Q = 4[\langle G^2\rangle - \langle G\rangle^2]}.$$

For M independent identical probes and an unbiased estimator the quantum Cramér–Rao inequality gives $\mathrm{Var}(\tilde{\phi}) \geq 1/(MF_Q)$.

Step 2: calculate for a coherent optical phase

The number operator generates an optical phase: $G = \hat{n}$. For $|\alpha\rangle$, $\langle n\rangle = |\alpha|^2 = \bar{n}$ and $\mathrm{Var}\,n = \bar{n}$, giving

$$\boxed{F_Q = 4\bar{n}, \quad \Delta\tilde{\phi} \geq \frac{1}{2\sqrt{M\bar{n}}}}.$$

This displays standard shot-noise scaling in total photon resource $M\bar{n}$. The factor of 2 depends on the parameter-generator definition, not on a fundamental change to the scaling.

Step 3: compare with actual measurement statistics

For one photon in a balanced Mach–Zehnder interferometer, write $P_1 = \cos^2(\phi/2)$, $P_2 = \sin^2(\phi/2)$. The classical Fisher information is

$$F = \sum_{j=1}^{2} \frac{(\partial_\phi P_j)^2}{P_j}.$$

Since $\partial_\phi P_1 = -\frac{1}{2}\sin\phi$, $\partial_\phi P_2 = +\frac{1}{2}\sin\phi$,

$$F = \frac{\sin^2\phi}{4}\left(\frac{1}{\cos^2(\phi/2)} + \frac{1}{\sin^2(\phi/2)}\right) = \boxed{1}$$

away from singular endpoints (with the continuous limit 1). The one-photon phase bound for M trials is $\Delta\phi \geq 1/\sqrt{M}$.

Physical check / common trap. The QFI is the optimal information available from a *state*; the classical Fisher information depends on which measurement is actually made. State the resource count.

Part III – Atomic physics

13 AT-01: Finite nuclear size in a hydrogen-like ion

Course source: 2026 Atomic Assignment 3 (`PHYS4055_A3.pdf`), Question 1(a).

Step 1: construct the perturbing potential

In atomic units ($a_0 = 1,\ e = \hbar = m_e = 1$), a point nucleus has potential $V_0(r) = -Z/r$. For uniform nuclear charge in a sphere of radius R,

$$V_R(r) = \begin{cases} -\dfrac{Z}{2R}\left(3 - \dfrac{r^2}{R^2}\right), & r < R, \\ -\dfrac{Z}{r}, & r \geq R. \end{cases}$$

The perturbation is therefore

$$\Delta V(r) = \begin{cases} Z\left(\dfrac{1}{r} - \dfrac{3}{2R} + \dfrac{r^2}{2R^3}\right), & r < R, \\ 0, & r \geq R. \end{cases}$$

It is nonnegative inside the nucleus: a distributed charge is less singular than a point charge.

Step 2: use first-order perturbation theory

For sufficiently small R, approximate an ns wavefunction as constant inside the nucleus. Its nucleus density for a hydrogenic state is $|\psi_{ns}(0)|^2 = Z^3/(\pi n^3)$. Then

$$\Delta E^{(1)}_{ns} \simeq 4\pi |\psi_{ns}(0)|^2 \int_0^R r^2 \Delta V(r)\, dr.$$

Do the integral explicitly:

$$\begin{aligned} \int_0^R r^2 \Delta V(r)\, dr &= Z \int_0^R \left(r - \frac{3r^2}{2R} + \frac{r^4}{2R^3}\right) dr \\ &= ZR^2 \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{10}\right) = \frac{ZR^2}{10}. \end{aligned}$$

Thus

$$\boxed{\Delta E_{\text{fns}}(ns) \simeq \frac{2Z^4R^2}{5n^3} = \frac{2Z^2}{5n^3}(ZR)^2 \quad \text{(Hartree)}}.$$

The result is positive and scales as R^2; it becomes negligible for a point nucleus. The radius R here is the *uniform-sphere* radius, $R = \sqrt{5/3}\, R_{\text{rms}}$.

Physical check and limitation

For $l > 0$, the wavefunction vanishes at the origin in the nonrelativistic central-field model, and the leading small-R correction is suppressed relative to s-states. For high-Z ions, relativistic wavefunctions and more accurate nuclear charge models may be needed.

> **Physical check / common trap.** Write the *difference* between finite- and point-nucleus potentials, not the full Coulomb energy. Use consistent atomic units and the correct radial measure $r^2 dr$.

14 AT-02: All terms for two equivalent d electrons

Course source: 2026 Atomic Assignment 3, Question 2; Week 8 angular-momentum lectures.

Step 1: count allowed antisymmetric states

A d electron has $l = 2$, with five m_l values and two spin projections: 10 spin-orbitals. Two equivalent electrons cannot occupy the same spin-orbital, so there are

$$\boxed{N_{\text{microstates}} = \binom{10}{2} = 45}$$

independent antisymmetric states. This count must equal the total degeneracy of the final terms.

Step 2: couple spatial and spin angular momenta

The orbital total can be $L = 0, 1, 2, 3, 4$; two spins $s = 1/2$ give $S = 0$ (antisymmetric singlet) or $S = 1$ (symmetric triplet). For two identical $l = 2$ electrons, exchanging their positions in the coupled orbital state contributes $(-1)^{2l-L} = (-1)^L$. Fermionic antisymmetry requires:

Orbital exchange	Spin exchange	Allowed L, S
L even (symmetric)	$S = 0$ (antisymmetric)	${}^1S,\ {}^1D,\ {}^1G$
L odd (antisymmetric)	$S = 1$ (symmetric)	${}^3P,\ {}^3F$

Step 3: attach the total electronic J

For each term, $J = |L - S|, \ldots, L + S$:

$$\boxed{\begin{array}{ll} {}^1S_0, & {}^1D_2, \quad {}^1G_4, \\ {}^3P_{0,1,2}, & {}^3F_{2,3,4}. \end{array}}$$

Check all degeneracies:

$$\underbrace{1}_{{}^1S} + \underbrace{5}_{{}^1D} + \underbrace{9}_{{}^1G} + \underbrace{(1+3+5)}_{{}^3P} + \underbrace{(5+7+9)}_{{}^3F} = \boxed{45}.$$

Step 4: apply Hund's rules with their caveats

First maximise total spin S, favouring triplets; next maximise L, favouring the ${}^3F$ term; since d^2 is less than half full, lowest J is favoured within that term:

$$\boxed{\text{Predicted ground term: } {}^3F_2}.$$

These rules do not uniquely order *all* excited terms. Consult the specified species in the NIST Atomic Spectra Database for the measured order.

> **Physical check / common trap.** Only listing $L = 0, \ldots, 4$ and $S = 0, 1$ overcounts forbidden states. Exchange antisymmetry and the 45-microstate check are essential.

15 AT-03: Hydrogenic Stark shift from closure

Course source: 2026 Atomic Assignment 3, Question 3(a)–(c).

Step 1: write the perturbation and parity argument

An atom in a uniform electric field $\mathcal{E}\hat{z}$ has perturbation $H_E = e\mathcal{E}z$ for an electron, because its dipole is $-e\mathbf{r}$. A non-degenerate $1s$ state has even parity, so $\langle 1s|z|1s\rangle = 0$ and the first-order Stark shift vanishes. In second order,

$$\Delta E_{1s}^{(2)} = \mathcal{E}^2 \sum_{m \neq 1s} \frac{|\langle m|ez|1s\rangle|^2}{E_{1s} - E_m} = -\frac{1}{2}\alpha\mathcal{E}^2.$$

Hence, in atomic units,

$$\boxed{\alpha = 2 \sum_{m \neq 1s} \frac{|\langle m|z|1s\rangle|^2}{E_m - E_{1s}}}.$$

Step 2: approximate all denominators

As requested in the assignment, take $E_m - E_1 \simeq E_2 - E_1 = 3Z^2/8$ Hartree. Using completeness,

$$\sum_{m\neq 1s} |\langle m|z|1s\rangle|^2 = \langle 1s|z^2|1s\rangle - |\langle 1s|z|1s\rangle|^2.$$

The second term is zero by parity. Spherical symmetry gives $\langle z^2\rangle = \langle r^2\rangle/3$. For the hydrogenic $1s$ orbital, $\langle r^2\rangle = 3/Z^2$ in Bohr-radius units. Therefore $\langle z^2\rangle = 1/Z^2$, and

$$\boxed{\alpha_{\text{approx}} = \frac{2(1/Z^2)}{3Z^2/8} = \frac{16}{3Z^4}\, a_0^3}.$$

Step 3: numerical checks and interpretation

For hydrogen, $Z = 1$:

$$\alpha_{\text{approx}} = \frac{16}{3}a_0^3 = 5.333\, a_0^3 \simeq 7.903 \times 10^{-25}\ \text{cm}^3,$$

using $a_0^3 = 1.48185 \times 10^{-25}$ cm^3. The exact $1s$ result is $\alpha_{\text{exact}} = 4.5\, a_0^3$, so this crude denominator approximation overestimates it by 18.5%.

For hydrogen-like Rb^{36+} $(Z = 37)$,

$$\boxed{\alpha_{\text{approx}} = \frac{16}{3(37)^4}a_0^3 = 2.846 \times 10^{-6} a_0^3 \simeq 4.217 \times 10^{-31}\ \text{cm}^3}.$$

The tightly bound single $1s$ electron is much less polarisable than valence electrons in *neutral* Rb.

> **Physical check / common trap.** Use parity before a long sum; in closure, exclude the initial state explicitly. The Z^{-4} scaling is a powerful plausibility test.

16 AT-04: Hyperfine structure, Landé factors and fields

Course source: 2025 PHYS4055 Exam, Question 1(a)–(e); Atomic Assignment 3, Question 4.

Step 1: find the hyperfine multiplets

For electronic term $^2P_{1/2}$, $J = 1/2$; let nuclear spin be $I = 5/2$. Coupling $\mathbf{F} = \mathbf{I} + \mathbf{J}$ yields

$$\boxed{F = |I - J|, \ldots, I + J = 2, 3}.$$

Using $F^2 = I^2 + J^2 + 2\mathbf{I}\cdot\mathbf{J}$ and $H_{\text{hfs}} = A\mathbf{I}\cdot\mathbf{J}/\hbar^2$,

$$\boxed{E_F = \frac{A}{2}\{F(F+1) - I(I+1) - J(J+1)\}}.$$

Since $I(I+1) = 35/4,\ J(J+1) = 3/4$,

$$\boxed{E_2 = -\frac{7A}{4}, \quad E_3 = \frac{5A}{4}, \quad E_3 - E_2 = 3A}.$$

For $A > 0$, $F = 3$ lies above $F = 2$; for $A < 0$, the order reverses.

Step 2: derive and evaluate g_J

With $g_L = 1,\ g_S \simeq 2$ and $\mathbf{J} = \mathbf{L} + \mathbf{S}$, projection of the electronic moment onto J gives

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}.$$

For $^2P_{1/2}$, $L = 1,\ S = 1/2,\ J = 1/2$:

$$\boxed{g_J = 1 + \frac{\frac{3}{4} + \frac{3}{4} - 2}{2(\frac{3}{4})} = \frac{2}{3}}.$$

Step 3: weak-field splitting

Ignoring the smaller nuclear magnetic moment,

$$g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}$$

for $F > 0$. Substitution gives

$$\boxed{g_{F=2} = -\frac{1}{9}, \qquad g_{F=3} = +\frac{1}{9}.}$$

At weak B, the energies are $E_{F,m_F} \simeq E_F + \mu_B g_F m_F B$. Thus $F = 2$ splits into five magnetic substates and $F = 3$ into seven; for generic weak nonzero B the 12 states have different energies. The slopes of the two manifolds have opposite signs.

Step 4: describe a strong field

When Zeeman energy dominates hyperfine coupling, the $|F, m_F\rangle$ basis is no longer approximately diagonal. Use $|Jm_J, Im_I\rangle$ instead:

$$E \simeq \mu_B g_J m_J B + A\, m_I m_J \quad \text{(neglecting nuclear Zeeman energy).}$$

There are two $m_J = \pm 1/2$ electronic branches, each with six m_I values. The electric-quadrupole hyperfine operator is rank 2; the electronic $J = 1/2 \to J = 1/2$ reduced matrix element vanishes because $J + J = 1 < 2$ (triangle rule).

> **Physical check / common trap.** A strong-field calculation must switch coupling basis, not simply extend the weak-field straight lines indefinitely. The $2F + 1$ degeneracies must sum to $(2I + 1)(2J + 1) = 12$.

17 AT-05: Hartree–Fock direct and exchange matrix elements

Course source: 2026 Atomic Assignment 4 (`PHYS4055_A2-3.pdf`), Question 1.

Step 1: antisymmetrise two occupied spin-orbitals

Let $a(x)$, $b(x)$ be orthonormal spin-orbitals; $x = (\mathbf{r}, \sigma)$. A two-electron determinant is

$$\Psi_{ab}(1,2) = \frac{1}{\sqrt{2}} \left[a(1)b(2) - b(1)a(2) \right].$$

The electron repulsion is $v_{12} = e^2/(4\pi\epsilon_0 r_{12})$. Expanding $\langle \Psi_{ab} | v_{12} | \Psi_{ab} \rangle$ gives four terms. Interchanging labels $1 \leftrightarrow 2$ shows that the two direct terms agree and the two exchange terms agree:

$$\boxed{\langle v_{12} \rangle = J_{ab} - K_{ab}}.$$

With atomic units, these integrals are

$$J_{ab} = \iint \frac{|a(x_1)|^2 |b(x_2)|^2}{r_{12}}\, dx_1 dx_2,$$
$$K_{ab} = \iint \frac{a^*(x_1) b(x_1) b^*(x_2) a(x_2)}{r_{12}}\, dx_1 dx_2.$$

The exchange term comes from fermionic antisymmetry; it is not a separate classical Coulomb force.

Step 2: integrate spins before orbital angles

For separable $a(x) = \phi_a(\mathbf{r})\chi_a(\sigma)$, $b(x) = \phi_b(\mathbf{r})\chi_b(\sigma)$, the direct term is independent of the relative spin. The exchange term contains $|\langle \chi_a | \chi_b \rangle|^2$. For orthogonal spin states α and β, this factor is zero: $\boxed{K_{a\alpha,b\beta} = 0}$. For equal spin states, K_{ab} generally survives.

Step 3: reduce an illustrative s-orbital pair

Take orbitals with $l_a = l_b = 0$ and $\phi_a(\mathbf{r}) = P_a(r)Y_{00}(\Omega)/r$ with $\int_0^\infty |P_a(r)|^2 dr = 1$. In the multipole expansion of $1/r_{12}$, only $k = 0$ survives the angular integrations. Write $r_> = \max(r_1, r_2)$. Then

$$\boxed{J_{ab} = \int_0^\infty \int_0^\infty \frac{|P_a(r_1)|^2 |P_b(r_2)|^2}{r_>} \, dr_1 dr_2}$$

and for equal spin

$$\boxed{K_{ab} = \int_0^\infty \int_0^\infty \frac{P_a^*(r_1) P_b(r_1) P_b^*(r_2) P_a(r_2)}{r_>} \, dr_1 dr_2}.$$

For higher l, retain the appropriate k, q angular coefficients (Gaunt/3-j factors).

Interpretation

The direct mean field is the Coulomb potential of the other electron density; exchange is a nonlocal, spin-selective operator. In Hartree–Fock, optimising the determinant self-consistently produces both terms in the one-electron equations.

Physical check / common trap. The full Assignment 4 derivation requires the Johnson-textbook angular and radial conventions. This example isolates the physical and algebraic direct–exchange mechanism, not every general k term.

Part IV – Applications of atomic physics

18 AA-01: Selection rules and radiative lifetime

Course source: Week 9 lectures; Week 10 spectroscopy; 2026 Assignment 4, Question 3 (sodium lifetimes).

Step 1: decide whether the matrix element can exist

For an E1 transition, the dipole operator d_q is a rank-1 tensor of *odd* parity:

$$\langle J'm'|d_q|Jm\rangle = (-1)^{J'-m'} \begin{pmatrix} J' & 1 & J \\ -m' & q & m \end{pmatrix} \langle J'\|d\|J\rangle.$$

Thus $m' = m + q$, $|J - J'| \leq 1 \leq J + J'$, and parity must change. For one electron in the LS-coupled limit, E1 orbital selection gives $\Delta l = \pm 1$. This predicts, for example,

$$3p \to 3s \text{ E1 allowed}, \quad 4s \to 3p \text{ E1 allowed}, \quad 3d \to 3p \text{ E1 allowed},$$

but $3d \to 3s$ is *not* E1 allowed because $\Delta l = -2$. It may be possible by higher multipoles. Whether a decay is physically available also requires the final state to lie below the initial energy.

Step 2: separate angular and radial factors

For $\psi_{nlm} = R_{nl}(r)Y_{lm}(\Omega)$, the dipole amplitude separates into

$$\langle f|r_q|i\rangle = \underbrace{\int_0^\infty R_f^*(r)R_i(r)r^3\,dr}_{\text{radial overlap}} \underbrace{\int Y_{l_f m_f}^*(\Omega)C_q^{(1)}(\Omega)Y_{l_i m_i}(\Omega)\,d\Omega}_{\text{angular / selection factor}}.$$

The reduced radial convention $u(r) = rR(r)$ uses $\int u_f^* r u_i\,dr$ instead. Never confuse R and u.

Step 3: convert an E1 dipole into a rate

For one specified transition with correctly summed polarisation components,

$$\boxed{A_{if} = \frac{\omega_{if}^3}{3\pi\epsilon_0\hbar c^3}|\langle f|\mathbf{d}|i\rangle|^2}, \quad \omega_{if} = \frac{E_i - E_f}{\hbar}.$$

If the excited state has several allowed lower states, $\Gamma_i = \sum_f A_{if}$ and $\boxed{\tau_i = 1/\Gamma_i}$. **Illustrative calculation (not measured Na data):** take a single 589 nm transition with $|\mathbf{d}| = 2ea_0$. Using $a_0 = 5.29177 \times 10^{-11}$ m and $\omega = 2\pi c/\lambda$ gives

$$A \simeq 3.97 \times 10^7 \text{ s}^{-1}, \qquad \boxed{\tau \simeq 25.2 \text{ ns}}$$

if no other decay channels contribute. For actual sodium $3p, 4s, 3d$, calculate the state-dependent dipole integrals and sum all E1-allowed downward channels as the assignment asks.

Step 4: contrast E1, M1 and E2

E1: rank 1, odd parity; M1: rank 1, even parity; E2: rank 2, even parity. Rank-2 angular momentum demands $|J - J'| \leq 2 \leq J + J'$. In particular, an $s \to d$ orbital transition can be E2, whereas $s \to p$ cannot be E2 by parity.

> **Physical check / common trap.** A nonzero 3-j symbol proves only angular compatibility. The reduced matrix element may still vanish; the full lifetime requires all permitted lower channels.

19 AA-02: Zeeman slowing and the Doppler limit

Course source: 2025 PHYS4055 Exam, Question 2(d)–(e); Week 10 tutorial `Tutorial_questions.pdf`, Question 2.

Given and modelling choice

Consider the 2025 Yb beam: $M = 1.73 \times 10^{-25}$ kg, source temperature $T = 300$ K, cooling wavelength $\lambda = 399$ nm, and excited-state lifetime $\tau = 5.5$ ns. The question does not define a unique initial velocity, so explicitly choose the 3D rms thermal speed $v_{\mathrm{rms}} = \sqrt{3k_BT/M}$. This gives

$$v_0 = \sqrt{\frac{3(1.38065 \times 10^{-23})(300)}{1.73 \times 10^{-25}}} = \boxed{268\ \mathrm{m\,s^{-1}}}.$$

The most probable speed $\sqrt{2k_BT/M}$ would give a different stopping distance; state which convention you use.

Step 1: maximum photon-scattering force

A saturated cycling transition can scatter at most approximately $\Gamma/2$ photons per second, $\Gamma = 1/\tau$. One absorbed photon transfers momentum $p_\gamma = h/\lambda$ opposite the atomic beam:

$$F_{\max} = \frac{h}{2\lambda\tau}, \quad a_{\max} = \frac{h}{2M\lambda\tau}.$$

The question specifies *half* the maximum deceleration:

$$\boxed{a = \frac{a_{\max}}{2} = \frac{h}{4M\lambda\tau} \simeq 4.36 \times 10^5\ \mathrm{m\,s^{-2}}}.$$

Step 2: stopping length and photon count

Using $v_f^2 = v_0^2 - 2aL$ with $v_f = 0$:

$$\boxed{L = \frac{v_0^2}{2a} \simeq 0.0823\ \mathrm{m} = 8.23\ \mathrm{cm}}.$$

Each counter-propagating photon changes the atom's momentum by h/λ, so the required net momentum Mv_0 corresponds to

$$\boxed{N_\gamma \simeq \frac{Mv_0}{h/\lambda} = \frac{Mv_0\lambda}{h} \simeq 2.79 \times 10^4\ \text{photons}}.$$

Spontaneous emission has zero *mean* momentum along the slowing axis but causes recoil heating.

Step 3: Doppler cooling and Zeeman resonance

The two-level Doppler cooling limit is

$$\boxed{T_D = \frac{\hbar\Gamma}{2k_B} = \frac{h}{4\pi k_B\tau} \simeq 6.94 \times 10^{-4}\ \mathrm{K} = 0.694\ \mathrm{mK}}.$$

In optical molasses, red-detuned counter-propagating beams are seen closer to resonance by an atom moving against them, creating velocity-dependent damping. As an atom slows, its Doppler shift changes; a Zeeman slower uses a position-dependent magnetic field to shift the atomic transition and maintain resonance. For approximately constant deceleration, $v(z) = \sqrt{v_0^2 - 2az}$, motivating a square-root magnetic-field profile when the Zeeman shift is roughly linear in B.

Physical check / common trap. Scattering force, slowing distance and Doppler temperature depend on different quantities. A small lifetime increases maximum force but also increases the simple two-level Doppler limit. Include the factor of 1/2 when the question asks for half maximum acceleration.

20 AA-03: Cesium clock splitting and weak-induced E1 amplitude

Course source: Week 13 clock slides; Week 12 atomic-parity-violation slides.

Example A: the clock's hyperfine states

For ^{133}Cs ground state $^2S_{1/2}$, $J = 1/2$ and nuclear spin $I = 7/2$. Therefore

$$\boxed{F = 3, 4}, \qquad I(I+1) = 63/4, \quad J(J+1) = 3/4.$$

Using the magnetic-dipole hyperfine formula,

$$E_3 = \frac{A}{2}\left(12 - \frac{63}{4} - \frac{3}{4}\right) = \boxed{-\frac{9A}{4}},$$
$$E_4 = \frac{A}{2}\left(20 - \frac{63}{4} - \frac{3}{4}\right) = \boxed{\frac{7A}{4}}.$$

The clock frequency is determined by

$$\boxed{E_4 - E_3 = 4A = h\nu_{\mathrm{Cs}}}, \qquad \nu_{\mathrm{Cs}} = 9\,192\,631\,770 \text{ Hz}.$$

Consequently $A/h = \nu_{\mathrm{Cs}}/4 \simeq 2.29816$ GHz. The actual clock transition is $|F = 3, m_F = 0\rangle \leftrightarrow |F = 4, m_F = 0\rangle$: both $m_F = 0$ states have no linear Zeeman shift in the weak-field approximation, reducing magnetic-field sensitivity. $\sum_F (2F+1) = 7 + 9 = 16 = (2I+1)(2J+1)$ checks the state count.

Example B: weak interaction opens a forbidden line

Suppose $|i\rangle$ and $|f\rangle$ have the *same* parity. An electric dipole d is parity-odd, therefore $\langle f|d|i\rangle = 0$ if parity is exact. Introduce a weak parity-violating perturbation H_W that mixes the initial state with an opposite-parity state $|n\rangle$:

$$|\tilde{i}\rangle \simeq |i\rangle + \epsilon\,|n\rangle\,, \qquad \epsilon = \frac{\langle n|H_W|i\rangle}{E_i - E_n}.$$

To first order,

$$\boxed{\langle f|d|\tilde{i}\rangle \simeq \epsilon\langle f|d|n\rangle}.$$

For an *illustrative* ratio $|\epsilon| = 10^{-8}$ and allowed $|\langle f|d|n\rangle| = 1\ ea_0$, the induced amplitude is only $10^{-8} ea_0$; a measurement needs a sensitive interference technique. In general, corrections to both initial and final states contribute, with a sum over intermediate states.

Why this connects to EDM experiments

An intrinsic permanent EDM along angular momentum would violate parity and time reversal. An external electric field gives $H_{\mathrm{EDM}} = -\mathbf{d}\cdot\mathbf{E}$. Field reversal is used to distinguish an EDM-like signal from many conventional shifts. Atomic structure and symmetry selection rules are thus essential for precision fundamental-physics tests.

Physical check / common trap. Do not confuse a *clock* transition frequency with the frequency of an allowed optical E1 line. Weak perturbations can produce tiny transitions that are exactly forbidden in the unperturbed parity basis.

Fast method finder

Recognise the question, then find its worked calculation

Question pattern	Example
Show displacement identity or coherent-state classical field	QO-01
Trace a TMSV state, calculate $g_{si}^{(2)}$, herald photon	QO-02
Transform Fock states, HOM or NOON interference	QO-03
Find photon variance or quadrature squeezing	QO-04
Partial trace a Bell pair, encode dense-coding messages	OA-01
Compute BB84 sifted key, QBER and eavesdropper detection	OA-02
Evaluate distillation success and improved fidelity	OA-03
Derive Kraus operators, reduced output and purity	OA-04
Normalise successful conditional filter	OA-05
Evolve two atoms in a common cavity and calculate populations	OA-06
Fourier solve a Langevin OPO and output squeezing spectrum	OA-07
Calculate quantum/classical Fisher information	OA-08
First-order finite-nuclear-size perturbation	AT-01
Equivalent-electron terms and microstate check	AT-02
Hydrogenic ground-state Stark polarizability by closure	AT-03
Hyperfine energies and weak/strong magnetic-field regimes	AT-04
Slater determinant direct vs exchange Coulomb integrals	AT-05
Apply E1/M1/E2 rules; compute a lifetime from dipole	AA-01
Max scattering force, stopping distance and Doppler limit	AA-02
Cesium clock hyperfine frequency; parity-mixed E1 amplitude	AA-03

Reusable five-step procedure: (1) identify the model and basis; (2) state operator/phase conventions; (3) calculate symbolically before substituting numbers; (4) normalise and check units, parity, state-count or probability; (5) interpret what is measured physically.